MILITARY SPECIFICATION

TEST SET, MISSILE CONTROL GROUP

OQ-50/AWM-21, OQ-51/AWM-21, and OQ-52/AWM-21

This Specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be for Organizational and Intermediate Level use and shall be comprised of two units. One unit shall be the Meter, Fault Isolation ME-373/AWM-21, ME-374/AWM-21, or ME-375/AWM-21 which is referred to as FIM. The other unit shall be the Alternating Current Null Detector (AC Null Detector) DT-345A/AWG-10. The FIM shall utilize a programmed tape which controls commands and permits comparator measurements of FIM inputs. The AC Null Detector shall be capable of measuring low voltage alternating current (ac) and direct current (dc) signals. The equipment is applicable to the AN/AWG-10, AN/AWG-11, and AN/AWG-12 Missile Control Systems (3.4.3). The equipments are interchangeable between all three systems except for certain cable assemblies that are carried in the FIM units.

1.2 Classification - The Missile Control Group Test Set shall consist of the following items:

Items	Applicable Paragraph
Fault Isolation Meter	3.4.1
Cable Assembly, Signal	3.4.1.9.1
Cable Assembly, Voltmeter Test Probe	3.4.1.9.2
Cable Assembly, Signal Adapter	3.4.1.9.3
Cable Assembly, Signal Adapter	3.4.1.9.4
Cable Assembly, Coaxial	3.4.1.9.5
Adapter, Connector	3.4.1.9.6

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Items	Applicable Paragraph
Cable Assembly, Signal Adapter	3.4.1.9.7
Adapter, Cable Assembly	3.4.1.9.8
Combination Case	3.4.1.10
Detector, Alternating Current Null DT-345A/AWG-10	3.4.2
Cable Assembly, AC Null Detector	3.4.2.4.7.1
Cable Assembly, AC Null Detector Signal	3.4.2.4.7.2
Combination Case	3.4.2.4.8

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-17	Cables, Radio Frequency: Coaxial, Dual Coaxial, Twin Conductor and Twin Lead
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Name plates for Aeronautical Electronic and Associated Equipment

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MIL-T-21200	Test Equipment for use with Electronic and Fire Control Systems, General Specification for
MIL-T-23336	Technical Directives (Letter Type) - Preparation of
MIL-P-26514	Polyurethane Foam, Rigid or Elastic, for Packaging
MIL-B-55130	Batteries, Storage, Sealed, Nickel-Cadmium
MIL-F-81526	Film-Tapes, Semiautomatic Test

STANDARDS

Military

MIL-STD-701	Lists of Standard Semiconductor Devices
MIL-STD-781	Reliability Test, Exponential Distribution

DRAWINGS

Military

MS3122	Connectors, Receptacle, Electric, Crimp-Type, Box Mounting, Flange, No. 4 Holes, Bayonet Coupling
MS3126	Connector, Plug, Electric, Crimp Type, Straight, Bayonet Coupling
MS24663	Connector, Plug, Electrical - with Spring-Loaded Pivoted Grounding Blade, Polarized
MS35183	Connector, Receptacle, Electrical, Series BNC, UG-912A/U

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2.1.1 Availability of Documents - Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting office. See 6.5.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(a) Parts and Materials as approved by MIL-T-21200 shall be given first consideration.

(b) Guidelines for the selection of nonstandard parts shall be in accordance with MIL-T-21200.

(c) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass electrical and environmental test.

3.1.1 Nonstandard Parts - Nonstandard parts and materials shall be as outlined in MIL-T-21200 except that mechanical parts shall be considered as nonstandard parts only when the material requirements of MIL-T-21200 as modified herein, are not met.

3.1.2 Tube Types and Semiconductor Types - The choice and application of electron tubes and semiconductor types used in the equipment shall be in accordance with MIL-T-21200 and MIL-STD-701.

3.1.3 NonCorrosive Hardware - External Military Standard (MS) hardware which shall be used during servicing and installing test equipments shall be of a corrosive-resistant type and shall not be further finish-treated.

3.2 Design and Construction - The equipment shall conform with all the applicable requirements of MIL-T-21200, as applicable for design, construction and workmanship, unless otherwise specified herein.

3.2.1 Total Weight - The total weight of the equipment, which is the FIM with its cables and the AC Null Detector with its cables, shall not exceed 70 pounds.

3.2.2 Reliability

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3.2.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently, for a period of 24 hours without necessity for readjustment of any operator controls which are inaccessible to the operator during use, except where adjustments associated with parts failures are required following replacement of those parts.

3.2.2.2 Operating Life - The FIM, and AC Null Detector shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. The AC Null Detector with fully charged batteries shall have a minimum operating life of 40 hours (3.3.2.4).

3.2.2.3 Mean Time Between Failures (MTBF) - When maintained in accordance with applicable preventive maintenance procedures, the equipment shall demonstrate a specified MTBF of at least 100 hours. Accept-reject criteria shall be as defined in paragraph 4.2.8.1 of MIL-STD-781.

3.2.3 Interchangeability - The equipment shall meet the interchangeability requirements as defined in MIL-T-21200 for each individual system.

3.2.4 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-T-21200 except that the FIM shall not be operated on its left or right sides.

3.2.5 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.2.6 Standard Conditions - The following conditions shall be used to estabish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient:	30° ±10° Centigrade (C) 86° ±18° Fahrenheit (F)
Altitude	Normal Ground	
Vibration	None	
Humidity	Room ambient up to 90% relative humidity	
Input Voltage	115 ±1.0 Volts Alternating Current (Vac), three-phase, 400 ±8 hertz (Hz) 26 ±1 Vac, single-phase, 400 ±8 Hz +27.5 ±1 Volts Direct Current (Vdc) +25.5 ±1 Vdc +6 ±1% Vdc -6 ±1% Vdc +12 ±1% Vdc +25 ±1% Vdc	

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-25 ±1% Vdc
-60 ±1% Vdc

Batteries, Internal — Six nickel - Cadmium, sealed, storage batteries

3.2.7 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class 2 equipment.

3.2.8 Warm-Up Time - The time required for the equipment to warm up prior to operations shall be kept to a minimum and shall not exceed 5 minutes under standard conditions, 20 minutes at the extreme service conditions.

3.2.9 Calibration - The equipment shall be calibrated at periodic intervals established on the basis of stability, purpose and degree of usage. Calibration intervals shall be adjusted as required to assure continued accuracy as evidenced by previous calibration results.

3.2.10 Primary Input Power Requirements - The equipment shall meet all the applicable requirements of MIL-T-21200 and shall meet Performance (3.3) and Detail Requirements (3.4) when supplied from the following power sources with characteristics as defined in MIL-T-21200 having limits as specified herein. In general, the Test Set shall be designed for operation over the frequencies as specified below. Operation below the specified frequencies shall result in no damage to the unit and the unit shall resume normal operation when the frequency returns within specified limits, provided the voltage decreases proportionately (zero voltage at zero frequency).

(a) 115 Vac
Operating Voltage Limits: 107.5 to 119.5 Vac, three-phase
Operating Frequency Limits: 380 to 420 Hz
Power Requirements: 100 Volt-Amperes (VA) maximum (max)

(b) +25 Vdc Protected
Operating Voltage Limits: 22 to 29 Vdc
Transients Limited to 43 Vdc max
Power Requirements: 175 watts max

(c) 28 Vdc
Operating Voltage Limits: 24 to 29 Vdc
Power Requirements: 100 watts max

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(d) 26 Vac
Operating Voltage Limits: 24 to 27 Vac, single-phase
Operating Frequency Limits: 380 to 420 Hz
Power Requirements: 30 VA max

(e) +6 Vdc
Operating Voltage Limits: 5.94 to 6.06 Vdc
Power Requirements: 300 milliwatts max

(f) -6 Vdc
Operating Voltage Limits: -5.94 to -6.06 Vdc
Power Requirements: 6 watts max

(g) +12 Vdc
Operating Voltage Limits: 11.88 to 12.12 Vdc
Power Requirements: 12 watts max

(h) +25 Vdc
Operating Voltage Limits: 24.75 to 25.25 Vdc
Power Requirements: 5 watts max

(i) -25 Vdc
Operating Voltage Limits: -24.75 to -25.25 Vdc
Power Requirements: 5 watts max

(j) -60 Vdc
Operating Voltage Limits: -59.4 to -60.6 Vdc
Power Requirements: 1 watt max

(k) 115 Vac (Battery Charging, AC Null Detector only)
Operating Voltage: 107.5 to 119.5 VAC, single-phase
Operating Frequency Limits: 50 to 420 Hz
Power Requirements: 10 VA max

3.3 Performance

3.3.1 Meter, Fault Isolation ME-373/AWM-21, ME-374/AWM-21 and ME-375/AWM-21 (FIM)

3.3.1.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under Service Conditions (3.2.7). When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

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3.3.1.2 Tape Control - First Level Address frames are those frames which can be programmable as a future address either from the Tape (3.3.1.3.1) in Automatic sequencing or from the MANUAL ADDRESS (3.4.1.3.(n)) thumbwheels in Manual tape control. There shall be 999 First Level Addresses. Second Level Addresses are the remainder of the frames on the Tape. These can be reached only by one step advancing in the forward direction in Automatic sequencing, or by half depressing FWD (Forward) (3.4.1.3.(a)) or REV (Reverse) (3.4.1.3 (b)).

(a) Automatic - The FIM shall progress from test to test on the Tape in a programmed fashion, stopping whenever a fault is located (Conditional Stop (3.3.1.3.3(a)), whenever an operator action (Unconditional Stop Pre-Test (3.3.1.3.3(b)) or decision (Unconditional Stop Post-Test (3.3.1.3.3(c)) is required, or at the completion of the tests. The operator shall be able to instruct the machine as to when an action has been completed by utilizing the GO (3.4.1.3(f))or No. GO(3.4.1.3(g)) controls. It may be necessary to program Not Fail if Exceed High (3.3.1.5(f)) and Not Fail if Exceed Low (3.3.1.5(g)) to allow manual decision inputs.

(b) Manual - The Tape may be programmed and moved to any First Level Address by setting the address on the MANUAL ADDRESS thumbwheels and depressing the appropriate FWD or REV button fully to the second detent position. The direction of Tape movement shall be determined by comparing the First Level Address number to the frame number in the visual Read Out Area 1 (3.3.1.4.1 (a)). When the FWD or REV button is depressed to the first detent position (approximately one-halfway depressed) the Tape shall move at a slower rate so that the frame numbers in the visual Readout Area 1 may be read. The Tape shall stop upon release of the button from the first detent position.

3.3.1.3 Test Sequence

3.3.1.3.1 Tape - The Test Set shall require a programmed Tape in order to operate. The Tape shall be per Film-Tapes, Semiautomatic Test, MIL-F-81526. The Tape shall contain information to control the output commands, information to control the FIM, information as to the test to be performed and test limits to be met, and, as necessary, other visual information required for the operator.

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3.3.1.3.2 Automatic Test Sequence - The Automatic Test Sequence shall be initiated by pressing the AUTO TEST (3.4.1.3(e)) button. The sequence shall be as follows:

(a) Tape selected output command lines shall be powered to drive relays.

(b) The signal to be examined shall be selected by the Tape program.

(c) The Comparator (3.3.1.7) shall be programmed as to voltage and tolerance which shall be acceptable. Any necessary signal modification circuits shall be called up.

(d) After an initial settling time, additional programmed output Delayed Command (3.3.1.11.1) lines may be driven and the programmed time delay Test Duration (3.3.1.11.2) shall be started.

(e) After the Test Duration delay, one of the following decisions shall be made based on the results of the test performed:

(1) Stop.

(2) Move one step to the next test on the Tape.

(3) Move to a programmed First Level Address test on the Tape.

(f) If the decision is Stop, the panel Meter (3.3.1.4.2) shall indicate the test results (3.3.1.4.2(a)) and Area 2 or Area 3 Readout shall be illuminated.

3.3.1.3.3 Programmed Stops - The Automatic Test Sequence (3.3.1.3.2) shall be interrupted by the following programmed commands and conditions.

(a) Conditional Stop - In the event Conditional Stop is programmed and the FIM has detected a failure, the sequence shall be terminated and the visual Readout Area 3 shall be illuminated.

(b) Unconditional Stop, Pre-Test - The test program shall stop after application of system control voltages but prior to the start of any timing sequences in the FIM.

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(c) Unconditional Stop, Post-Test - The test program shall stop after the timing sequence is complete on a particular frame.

3.3.1.3.4 Manual Test Sequence - The Manual Test Sequence shall be initiated by pressing the MAN TEST (3.4.1.3(c)) button. If AUTO TEST (3.4.1.3(e)) has been pressed it shall be necessary to press STOP (3.4.1.3(d)) in order to bring the FIM out of the Automatic Test Sequence (3.3.1.3.2) before it can be put into the Manual Test Sequence. In the Manual Test Sequence, steps 3.3.1.3.2(a) through 3.3.1.3.2(d) shall be duplicated. The difference shall be that no Tape action shall be permitted. Readout shall be by the Meter and visual readout area.

3.3.1.4 Indication

3.3.1.4.1 Readout - A three area illuminated visual Readout shall be provided.

(a) Area 1 - This area shall be visible at each tape stop. The area shall contain continuously illuminated information as to Tape frame number and name of test that is being performed.

(b) Area 2 and 3 - These two areas shall be viewed individually when the Tape is stopped. These areas shall contain information on results of test or actions required of the operator. Area illumination shall be determined by a combination of test results and the Tape program.

3.3.1.4.2 Meter - A panel Meter shall be provided having both left zero and center zero markings. Test limit zones shall be provided. One zone shall be centered about the mid scale zero position, and the left and right extremes shall comprise two additional zones.

(a) The Meter shall read from center zero when in the Comparator (3.3.1.6.1) Mode of operation. The indication displayed shall be the error from the Tape (3.3.1.3.1) programmed nominal value.

(b) The Meter shall read from left zero when in Voltmeter (3.3.1.6.2) Modes DC+, DC-, and AC.

(c) The Meter shall read from center zero when in the Voltmeter NULL (3.3.1.6.2(c)) Mode and shall not limit a voltage in the negative direction at any value more positive than -1 Vdc.

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3.3.1.4.3 Test in Progress - There shall be a light indication 320 ±80 milliseconds after the Automatic or Manual Test Sequence (3.3.1.3.2, 3.3.1.3.4) is initiated and shall remain illuminated for the Test Duration. The light indication shall be amber in color and shall be located in the upper portion of the front panel.

3.3.1.5 Programmed Commands - The following control commands shall be capable of being programmed on the tape.

(a) Signal Source - These instructions shall identify and command the necessary control lines to select any signal line from the FIM #1 (Fault Isolation Meter #1) (3.4.1.4(e)) connector 51J1, or the signal on the BIT #1 (Built-in Test #1) (3.4.1.4(c)) connector 51J3.

(b) Signal Character - A command shall cause activation of either the ac, dc, or pulse measurement circuitry.

(c) Signal Polarity - Commands shall program measurement circuitry for dc negative inputs or for ac asynchronously demodulated signals for signals, falling between 90° and 270° out of phase with respect to the reference.

(d) Nominal Signal Voltage - Necessary commands shall be sent to the comparison circuit to set it for the dc voltage about which acceptance limits are to be established.

(e) Allowable Signal Tolerance - Acceptance limits, as a percentage of the Nominal Signal Voltage (3.3.1.5(d)) shall be set as electronic trip points.

(f) Not Fail if Exceed High - Commands shall be programmed for elimination of the fail high limit on a high input signal.

(g) Not Fail if Exceed Low - Commands shall be programmed for elimination of the fail low limit on a low input signal.

(h) Peak Detect - A command shall cause insertion of the peak detection circuitry.

(i) Peak Detect, AC Coupled - A command shall be programmed to discriminate between pulse waveforms and dc voltages.

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(j) AC Asynchronous Detection - A programmed command shall cause an ac measurement to be made based on asynchronous detection with respect to a phase reference.

(k) Alternate AC Reference - A command shall change the AC phase reference from one point to a second point.

(l) Ground Reference Selection - Commands shall enable the selection of alternate dc measurement reference points.

(m) Offset as Reference - A command shall select a front panel adjustable dc Offset (3.4.1.3(i)) as a reference for dc measurements.

(n) Ground Presence Check - A command shall permit verification of impedance to ground.

(o) Test Time Duration - Programmed commands shall select one of four Test Duration delays.

(p) System Command to Terminate Test - A command shall permit a system signal to override the Test Duration delay provided the signal occurs before the end of the programmed Test Duration.

(q) Future Address Location - A First Level Address coded on the Tape shall be commanded to transfer from the Tape to the memory register for storage and subsequent address comparison with Tape frame addresses.

(r) Immediate Commands to System - A group of commands to the system shall be programmable in any combination.

(s) Delayed Commands to System - A group of Delayed Commands shall be delayed from the Immediate Commands to System and shall be programmable in any combination.

(t) Digital-to-Analog Voltage Out - Tape commands shall be programmed to provide a Digital-to-Analog Output (3.3.1.12) voltage in discrete steps from 0 to ±5.6 Vdc maximum which shall be connected to BIT #2 (Built-in Test #2) (3.4.1.4(d)) connector 5IJ2.

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(u) Polarity Change of Digital-to-Analog Voltage Out - A Tape command shall change the voltage polarity from normally negative to positive.

(v) Manual BIT Panel Select - A command shall cause the selection of specific, manually inserted, signals in the Manual Built-in-Test Panel of the Radar Maintenance Table-Rack to be sent to the system.

3.3.1.6 Modes of Operation - The FIM shall be capable of operating as either a Comparator or as a Voltmeter. These two Modes of Operation shall be selected in accordance with the following by the MODE (3.4.1.3(k)) control:

3.3.1.6.1 Comparator - This mode shall be the normal tape programmed operation. In the NORM (Normal) switch position, the Meter shall read as a center zero scale meter.

3.3.1.6.2 Voltmeter - In the Voltmeter Mode of operation, the MODE control shall select the following positions:

(a) DC+ - The FIM shall be used as a dc Voltmeter to measure inputs more positive than ground.

(b) DC- - The FIM shall be used as a dc Voltmeter to measure inputs more negative than ground.

(c) NULL - The FIM shall be used as a dc Voltmeter with a center zero scale for inputs of either polarity. There shall be limiting at -1 V for negative voltages when on the center zero scale.

(d) AC - The FIM shall be used as an ac Voltmeter to measure voltages from 0.02 to 250 Vac.

3.3.1.7 Comparator - The equipment shall have a tape programmed voltage Comparator with the following capability.

3.3.1.7.1 Frequency - The Comparator shall operate at dc and at ac frequencies from 0.5 to 8000 Hz.

3.3.1.7.2 Accuracy - The accuracy of the Comparator shall be as follows:

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3.3.1.7.2.1 DC

(a) 100mV to 3.330V: ± [0/04 (% SPAN PROG) + 0/16% + (1.6mV/v_n) 100%]

(b) 3.340 V to 24.99 V: ± [0.04 (% SPAN PROG) + 0.178% + (1mV/v_n) 100%]

(c) 25.0 V to 33.4 V: ± [0.04 (% SPAN PROG) + 0.39% + (1.6mv/V_n) 100%]

(d) 33.5 V to 250 V: ± [0.04 (% SPAN PROG) + 0.368% + (1mv/V_n) 100%]

Where: SPAN PROG is program tolerance in % and V_n is nominal voltage.

3.3.1.7.2.2 AC Synchronously Demodulated, or Average Detected

(a) 360 to 440 Hz and 810 to 990 Hz

1. 0.02 to 25 V rms, ±1% tolerance
2. 25 to 250 V rms, ±5% tolerance

(b) 60 to 70 Hz, 0.5 to 250 V rms, ±5% tolerance

3.3.1.7.2.3 Positive and Negative* Peak Voltage Detected

(a) 0.5 to 3 V, ±10% tolerance
(b) 3 to 35 V, ±5% tolerance

3.3.1.7.3 Resolution - The Comparator shall have the following resolution:

3.3.1.7.3.1 DC

Volts	Resolution
0.1 to 25 Vdc	0.01 Vdc
25 to 250 Vdc	0.1 Vdc
250 to 300 Vdc	0.3 Vdc

3.3.1.7.3.2 AC

Vrms	Resolution
0.02 to 1 Vac	0.002 Vac
1.0 to 10 Vac	0.01 Vac
10 to 100 Vac	0.1 Vac
100 to 250 Vac	1.0 Vac

*Negative peak detection applies to ECP 46 configuration only, which is pending Navy approval of nomenclature.

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3.3.1.7.3.3 Positive Peak

Volts (Peak)	Resolution
0.5 to 35 V	0.01 V

3.3.1.7.4 Voltage Range - The Comparator shall have the following voltage ranges:

3.3.1.7.4.1 DC - 0.1 to 300 Vdc

3.3.1.7.4.2 AC

(a) 360 to 440 Hz and 810 to 990 Hz: 0.02 to 250 Vac

(b) 60 to 70 Hz 0.5 to 250 Vac

3.3.1.7.4.3 Positive Peak - 0.5 to 35 V

3.3.1.7.5 Test Limits AC and DC - Test acceptance limits of ±1% to ±15% programmable in 1% steps shall be provided.

3.3.1.7.6 Protection AC and DC- Necessary protection shall be provided so that no system over-voltage on the signal cable, up to 330 Vdc or 250 Vac shall damage the Comparator on any range.

3.3.1.7.7 Impedance - The following minimum (min) input impedance shall be provided for the specified ranges:

(a) DC - 0.1 to 5.0 V: 30 Megohms
5.0 to 25 V: 30 Megohms
25 to 300 V: 1.0 Megohm

(b) AC - Minimum of 100 kilohms at 900 Hz on all ranges with Signal Cable Assembly (3.4.1.9.1) connected.

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(c) Peak Detection - 100,000 ohms minimum

3.3.1.7.8 AC Detector - When reading ac voltages, the Comparator shall operate either as a synchronous demodulator with a 400 or 900 Hz signal phase reference obtained from the unit under test, or as an average detector. The output when operating as an average detector shall be calibrated in terms of the equivalent rms value of a sine wave.

3.3.1.8 Voltmeter - The FIM shall be capable of use as a high impedance Voltmeter to monitor any selected inputs including those from an external test probe.

3.3.1.8.1 Voltage Range - The front panel Range (3.4.1.3(j)) switch shall indicate the following full scale ac and dc volts: 0.1, 0.3, 1, 3, 10, 30, 100, and 300. The 300 position shall be used for 300 Vdc or 250 Vac

3.3.1.8.2 Frequency - The FIM shall measure voltages at dc, and at ac frequencies from 60 to 990 Hz.

3.3.1.8.3 Accuracy - The Voltmeter accuracy shall be:

(a) dc - 0.1 to 300 V	±3% of Full Scale
(b) ac - 360 to 440 Hz and 810 to 990 Hz	
0.02 to 30 V	±3% of Full Scale
30 to 250 V	±6% of Full Scale
(c) ac - 60 to 70 Hz, 0.02 to 250 V	±6% of Full Scale

3.3.1.8.4 Impedance - The following minimum input impedance shall be provided for the specified ranges:

(a) dc - 0.1 to 3.0 V	33 Megohms
3.0 to 25 V	100 Megohms
25 to 300 V	1.0 Megohm

(b) ac - 100 kilohms at 900 Hz on all ranges with the Signal Cable Assembly (3.4.1.9.1) connected.

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3.3.1.9 Signal Inputs - The FIM shall receive inputs through the Signal Cable Assembly, the Connector Adapter, the Power Cable Assembly of either the Cable and Adapter Group, OA-8124/AWM-21 or the Cable Assembly Group, OA-8127/AWM-22 or the Voltmeter Test Probe Cable Assembly.

3.3.1.9.1 Signal Cable Assembly Inputs - The Signal Cable Assembly shall mate directly with a 50-pin connector of the particular Line Replaceable Unit (LRU) under test or shall connect via a Connector Adapter to a 25-pin connector of the LRU under test. As many as 48 test signals the Signal Cable Assembly. These test signals shall transfer to the FIM through FIM #1 connector, 51J1.

3.3.1.9.2 Power Cable Assembly Inputs - The Power Cable Assembly shall mate with the Selector-Test Programmer, SA-1447/APG-59, and provide the following inputs to the FIM through BIT #1 and BIT #2 connectors, 51J2 and 51J3:

(a) ac power

(b) dc power

(c) System selected signals

(d) Range voltages

3.3.1.10 Signal Outputs - The FIM provides outputs via the Signal Cable Assembly, the Connector Adapter, and the Power Cable Assembly of either the Cable and Adapter Group or the Cable Assembly Group.

3.3.1.10.1 Signal Cable Signal Transfer Capability - Two signal pins of the FIM #1 connector shall be capable of being jumpered to connect four alternate signal pins of the same connector. These pins shall transfer signals originating in the LRU under test and shall afford sixteen combinations of signal transfer.

3.3.1.10.2 Signal Cable Assembly Outputs - Digital to Analog Output voltages shall be capable of being programmed, through a 15,000 ohm resistor, to appear on any one of ten different pins of FIM #1 connector.

3.3.1.10.3 Power Cable Assembly Outputs - The FIM shall provide 61 output commands of the following types to the system.

(a) A BIT OFF interlock signal shall be provided during FIM operation.

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(b) A +28 Vdc system relay latch command shall be provided.

(c) A FIM clear command shall be provided.

(d) A Digital to Analog voltage, on command, shall be provided.

(e) A +26 Vdc Protected On-Off command shall be provided.

(f) Sixty-one + 26 Vdc commands shall be provided from the sensors. Some commands may be Delayed Commands.

3.3.1.11 Delays

3.3.1.11.1 Delayed Commands - A total of twenty programmed commands of the 61 sensor commands (3.3.1.10.3(f)) shall be delayed 320 ±80 milliseconds from the start of the test.

3.3.1.11.2 Test Duration - The following delays, from the end of the Delayed Commands to the initiation of the GO/NO-GO decision, shall be programmed from the Tape:

(a) 1 ±0.03 seconds

(b) 3 ±0.09 seconds

(c) 6 ±0.18 seconds

(d) 10 ±0.3 seconds

3.3.1.12 Digital to Analog Output - The FIM shall be capable of supplying the following dc output voltages either directly to BIT #2 connector or to FIM #1 connector through a 15,000-ohm resistance. The output voltages shall be programmed from the tape.

(a)	Voltage Range	0 to ±5.6 Vdc
(b)	Resolution	0.8 Vdc
(c)	Accuracy	±2%
(d)	Source Impedance	1000 ohms max when provided to BIT #2 connector. From 16,150 ohms max to 14,850 ohms min shall be the range when provided to FIM #1 connector.

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3.3.1.13 Sensors

3.3.1.13.1 Layout - There shall be 144 light sensitive transistor sensors provided. Positioning of the sensors shall be compatible with the Tape and its physical configuration.

3.3.1.13.2 Function - Function of the Sensors shall be to control the FIM and its Signal Outputs (3.3.1.10). Sensors shall be controlled by the Programmed Commands or the front Panel Controls.

3.3.2 Detector, Alternating Current Null (AC Null Detector) DT-345A/AWG-10

3.3.2.1 General - The AC Null Detector shall consist of a bandpass amplifier and meter Indicator (3.4.2.4.5) with the following characteristics:

3.3.2.2 ac Operation

(a) Bandwidth: 250 Hz min to 650 Hz max

(b) Sensitivity (for full scale deflection):

(1) Low Sensitivity: 3.5 to 4.5 millivolts, rms

(2) High Sensitivity: 250 to 400 microvolts, rms

3.3.2.2.1 Minimum Discernible Signal - The minimum discernible signal with High Sensitivity shall be 50 microvolts or less.

3.3.2.3 DC Operation - The signal input for full scale deflection of the meter shall be 3.0 Vdc ±10%.

3.3.2.4 Operating Life - After recharge of the batteries, at an ambient of 0°C (32°F) but not more than 65°C (149°F) for a period not exceeding 15 hours, the AC Null Detector shall perform within specification for a period of not less than 40 hours.

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3.4 Detail Requirements - The Missile Control Group Test Set shall meet the following requirements.

3.4.1 Meter, Fault Isolation ME-373/AWM-21, ME-374/AWM-21 and ME-375/AWM-21 (FIM)

3.4.1.1 Function - The FIM shall be capable of following the instructions programmed on the Tape (3.3.1.3.1) and shall be capable of performing dc, ac, and pulse measurements. The FIM shall make a decision based on the results, move the tape according to the results and shall indicate to the operator the results and instructions either dependent or not dependent on the results.

3.4.1.2 Form Factor - The dimensions of the FIM less cover shall not exceed 16 inches by 13 inches by 12 inches. The total weight including the Combination Case (3.4.1.10) and the Cables (3.4.1.9) shall be 58 pounds max. The Test Set less cover and accessories shall be 45 pounds max.

3.4.1.3 Front Panel Controls - The front panel shall contain the following:

(a) FWD (Forward) - This control shall be a two-position pushbutton that will enable the operator to manually move to any test frame on the program tape. The first detent position will advance the tape at 3 frames per second in the forward direction. Pressing this control to the second detent and releasing will automatically move the tape in the forward direction, at Slewing Speed at 10 frames per second to the address encoded by the MANUAL ADDRESS (3.4.1.3(n)). This pushbutton shall be blue in color.

(b) REV (Reverse) - This control shall be a two-position pushbutton that will enable the operator to manually move to any test frame on the program tape. The first detent position will advance the tape at 3 frames per second in the reverse direction. Pressing this control to the second detent and releasing will automatically move the Tape in the reverse direction, at Slewing Speed at 10 frames per second to the address encoded by the MANUAL ADDRESS. This pushbutton shall be blue in color.

(c) MAN TEST (Manual Test) - This button shall initiate the Manual Test Sequence. This pushbutton shall be blue in color.

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(d) STOP - This shall be a pushbutton control to stop the sequence of the FIM when it is progressing in any mode. This pushbutton shall be red in color.

(e) AUTO TEST (Automatic Test Sequence) - This shall be a pushbutton control to select the Automatic Test Sequence. This pushbutton shall be blue in color.

(f) GO - This shall be a control to manually encode a GO decision into the programmer logic circuits. It shall indicate to the logic circuits that a decision has been made or an action has been completed. The test sequence shall one step advance upon operation of the GO control. This pushbutton shall be blue in color.

(g) NO-GO - This shall be a control to manually encode a NO-GO decision into the programmer logic circuits. It shall indicate to the logic circuits that a preprogrammed fault isolation test shall be performed. The test sequence shall seek a future address upon operation of the NO-GO control. This pushbutton shall be blue in color.

(h) LOW SENS (Low Meter Sensitivity) - This control shall desensitize the Meter in the Comparator Mode of operation. It shall add 20% to 40% to the percentage which has been programmed as full scale voltage on the Meter (e.g., thus permitting 2% full scale to be desensitized from 22% to 42% of programmed voltage). This pushbutton shall be red in color.

(i) OFFSET (Offset Voltage Adjustment) - This shall be a control to manually offset the Meter from 0 to $\pm$6 V min. It shall be employed when the value to be measured is referenced to an initial reading and shall be enabled by the tape program.

(j) RANGE - This shall be a rotary selector switch to select the desired full scale deflection in the Voltmeter (3.3.1.6.2) Mode of operation only. AC and dc full-scale voltages shall be marked 0.1, 0.3, 1, 3, 10, 30, 100, and 300. In position 300, 300 Vdc and 250 Vac shall be measured.

(k) MODE - This switch shall select the Modes of Operation of the FIM and the Meter. The positions shall be marked NORM, DC+, DC-, NULL, and AC.

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3.3.1.3 Programmed Commands - The following control com-

(l) FIM (Fault Isolation Meter Enable) - This shall be a push-button operated switch-light which shall indicate operation of the FIM. Pushing the switch shall invert the operation of the FIM and the Selector-Test Programmer. This push-button indicator shall be red.

(m) BIT (Built in Test Enable) - This shall be a pushbutton operated switch-light which shall indicate operation of the Selector-Test Programmer. Pushing the switch shall invert the operation of the FIM and the Selector-Test Programmer. This pushbutton indicator shall be red.

(n) MANUAL ADDRESS - The FIM shall have a set of three digit, lighted thumbwheels which shall encode First Level Addresses (3. 3. 1. 2) into the FIM control logic circuitry.

(o) PANEL LIGHTS - A toggle switch which, when moved from OFF to ON, shall illuminate all the pushbuttons except BIT and FIM. It shall also illuminate the panel Meter and the MANUAL ADDRESS, MODE, and RANGE switches.

3. 4. 1. 4 Front Panel Connectors - There shall be six connectors on the front panel. Connectors shall be identified and shall be:

(a) OUTPUT - A MS35183 connector (51J5) shall be provided to monitor the input to the voltage measurement circuit.

(b) PROBE - A MS35183 connector (51J22) shall be provided for the external Voltmeter Test Probe Cable Assembly. A warning notice shall be adjacent to the connector having the following: CAUTION MAX VOLTS 330 VDC, 250 VAC.

(c) BIT #1 (Built in Test #1) - A connector (51J3) shall be provided for the Power Cable Assembly of the Cable and Adapter Group or the Cable Assembly Group to connect the Selector-Test Programmer and the FIM.

(d) BIT #2 - A connector (51J2) shall be provided for the Power Cable Assembly of the Cable and Adapter Group or the Cable Assembly Group to connect the Selector-Test Programmer and the FIM.

(e) FIM #1 (Fault Isolation Meter #1) - A connector (51J1) shall be provided for the Signal Cable Assembly to connect the Test Set and the unit under test.

(f) INPUT - A MS35183 connector (51J4) shall be provided to permit insertion of external voltages.

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3.4.1.5 Lighting

3.4.1.5.1 Indicator Lights - The following front panel light displays shall be provided.

(a) FIM - A red indicator shall be illuminated by a clear lamp when the FIM has control of the system.

(b) BIT - A red indicator shall be illuminated by a clear lamp when the Selector-Test Programmer has control of the system.

(c) Test In Progress - An amber indicator shall be illuminated by a clear lamp when the Automatic or Manual Test Sequence has been initiated. The light shall remain illuminated for the Test Duration.

3.4.1.5.2 Front Panel Lighting - The front panel shall be illuminated per MIL-T-21200. All light bulbs shall be replaceable from the front panel with the exception of the Panel Meter (3.4.1.6.1) and the Manual Address thumbwheel lights

3.4.1.6 Meters

3.4.1.6.1 Panel Meter - The Panel Meter shall be a 3-1/2 inch, center and left zero scale reading type, marked with GO and NO-GO zones. The reading scale shall have an arc of 100 degrees. A zone of $\pm 25^{\circ}$ centered on the center zero position shall be marked in green and shall represent a GO test measurement. The extreme end zones of 25° each shall be red and shall represent a NO GO test measurement. The scale shall also have appropriate voltage markings and shall be internally illuminated.

3.4.1.6.2 Elapsed Time Meter - The FIM shall contain a time totalizing meter capable of registering 9,999 hours as a minimum reading. The meter shall be located on the front panel and shall conform to MIL-M-7793.

3.4.1.7 Tape Readout Area - The Tape Readout Area shall consist of a window which shall permit the operator to view the program Tape instructions. The window shall have an effective Readout Area of not less than 3 square inches.

3.4.1.7.1 Magnifier - A fiber optics magnifier shall provide the means of magnifying information contained on the Tape and viewed in the window. The magnifier shall have a magnification of at least 2X.

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3.4.1.8 Tape Transport - The FIM shall contain a Tape Transport frame reader with characteristics as follows.

3.4.1.8.1 Program Tape - The Tape Transport shall be compatible with the program Film Tape Semiautomatic.

3.4.1.8.2 Tape Capacity - The Tape Transport shall be capable of accommodating a Tape at least 3, 100 inches in length.

3.4.1.8.3 Slewing Speed - The Tape Transport shall have a minimum speed of 10 frames per second at full-action speed and 2 to 4 frames per second a half-action speed.

3.4.1.8.4 Protection of Tape - Inadvertent operation of the Front Panel Controls shall not cause destruction of either the Tape or the Tape Transport even if at the end of a Tape reel.

3.4.1.9 Cables - One each of the following Cable Assemblies and Connector Adapters shall be required.

3.4.1.9.1 Cable Assembly, Signal (51W2) - This Cable Assembly shall couple AN/AWG-10/11/12 system unit test connectors to the FIM either directly or through a Connector Adapter. One end of the Cable Assembly shall provide a nominal 50 pin connector and the other end shall provide a socket connector, to mate with the FIM #1 connector of the FIM. The wire shall be shielded. The Cable Assembly shall be 103 ±6 inches in length.

3.4.1.9.2 Cable Assembly, Voltmeter Test Probe (51W3) - This Cable Assembly shall provide means of directly coupling AN/AQG-10/11/12 test points or test signals to the FIM. One end shall be a probe and a clip. The other end shall be a UG-260/U connector. A coaxial cable RG-210/U per MIL-C-17/97A shall connect the two ends. A 14 ±1-inch wire lead shall connect the clip to the outer conductor of the coaxial cable at the cable entry end of the probe. The Cable Assembly shall be 60 ±2 inches in overall length.

3.4.1.9.3 Cable Assembly, Signal Adapter (51W4) - This Cable Assembly shall be an adapter to 3.4.1.9.1 and shall also provide coupling of three AN/AWG-10 system unit test connectors. One end of the Cable Assembly shall have a single connector and the other end shall have three connectors. The cables from each of the three connectors shall be of equal length and shall junction as close as possible to the single connector. The cable assembly shall be 26 ±2 inches in length.

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3.4.1.9.4 Cable Assembly Signal Adapter (51W5) - This Cable Assembly shall be an adapter to Signal Cable Assembly and shall provide coupling of four AN/AWG-10 and AN/AWG-11 system unit test connectors. One end shall be a single connector and the other end shall have four connectors. The cables connecting each of the four connectors shall be of equal length and shall junction as close as possible to the single connector. The cable assembly shall be 26 ±2 inches in length.

3.4.1.9.5 Cable Assembly, Coaxial (51W6) - This Cable Assembly is used to couple two points on the AN/AWG-10/11/12 Radio Frequency Oscillator which is part of the equipment to be tested by the FIM. Both ends shall have a type TSM connector. The cable shall be coaxial per MIL-C-17/70 with center conductor. The Cable Assembly shall be 12 ±1 inch in length.

3.4.1.9.6 Adapter, Connector (51W6 and 55W2) - This Adapter is used to connect 25 pin AN/AWG-10/11/12 LRU connectors to the 50-pin Signal Cable Assembly connector. On each end there shall be a single connector. The Adapter shall be 8-3/4 ± 1/8 inches in length.

3.4.1.9.7 Cable Assembly, Signal Adapter (51W4) - This Cable Assembly shall be an adapter to the Signal Cable Assembly and shall also provide coupling of three AN/AWG-12 only system unit test connectors. One end of the Cable Assembly shall have a single connector and the other end three connectors. The cables from each of the three connectors shall be of equal length and shall junction as close as possible to the single connector. The cable Assembly shall be 26 ±2 inches in length.

3.4.1.9.8 Adapter Cable Assembly (51W7) - This Cable Assembly shall be an adapter to Signal Cable Assembly and shall provide coupling of eight AN/AWG-12 only system unit test connectors. One end shall have a connector box with two connectors and the other end shall have eight connectors. The cables to each of the eight connectors shall be of equal length and shall junction in the connector box. The Cable Assembly overall length shall be 26 1/2 ± 1/2 inches.

3.4.1.9.9 Cable Assembly, Power - This Cable Assembly shall be required to connect all necessary power to the FIM and carry the output commands to the Selector-Test Programmer. This Cable Assembly shall not be provided as part of the FIM but shall be supplied with the Cable and Adapter Group, OA-8124/AWM-21 and the Cable Assembly Group, OA-8127/AWM-22.

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3.4.1.10 Combination Case - The FIM shall be provided with a Combination Case which shall conform to MIL-T-21200. In addition it shall have adapter blocks for mounting on the Radar Maintenance Table Rack.

3.4.1.10.1 Size - The overall dimensions, including all protrusions such as handles, latches, latch guards, feet, and adapter blocks including all protrusions for mounting shall not exceed 13 1/2 inches by 15 1/2 inches by 17 3/4 inches.

3.4.1.10.2 Cover - Provision shall be made in the cover to store the Cable Assemblies and Adapters (3.4.1.9.1) through (3.4.1.9.8).

3.4.2 Detector, Alternating Current Null DT-345A/AWG-10

3.4.2.1 Function - The AC Null Detector shall be a hand-held meter for measuring low voltage ac and dc signals.

3.4.2.2 Form Factor - The overall dimensions of the AC Null Detector less cover shall be 7-1/2 inches by 5 inches by 6 inches. The total weight of the unit shall not exceed 13 pounds.

3.4.2.3 Power Requirements - The equipment shall receive necessary power to operate be means of six internal, series connected, rechargeable batteries. The total voltage shall be between 7.0 and 9.0 Vdc. External power of the following type shall be required for recharging the batteries (3.4.2.4.1):

(a) AC Power: Single-phase, 107.5 to 119.5 V, 50 to 420 Hz, not to exceed 10 VA max.

3.4.2.4 Contents of Unit - The AC Null Detector shall consist of the following:

3.4.2.4.1 Batteries - Batteries shall be provided to supply dc power to the AC Null Detector. Six, nickel-cadmium, sealed, storage Batteries equivalent to MIL-B-55130 standards, shall be furnished. Each Battery shall have a diameter not to exceed 0.6 inch and a length not to exceed 2.0 inches. Nominal open circuit voltage of each shall be 1.25 Vdc with a nominal capacity of 470 milliampere-hours at a 1 hour discharge rate and of 510 milliampere-hours at a 5-hour discharge rate. Battery terminal polarities shall be negative at the case and positive at the button. Metallic tabs shall be attached to the terminals to provide means of permanently soldering the batteries to the electrical circuit. Batteries shall be rechargeable at a constant charging current between 20 and 60 milliamperes (MA).

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3.4.2.4.2 Battery Charging Circuit - A Battery charging-circuit shall be provided that shall be capable of supplying a minimum of 20 MA to completely discharged batteries. This circuit shall be selected only in the BATT CHG (3.4.2.4.3(a)(2)) position of the Function Switch (3.4.2.4.3.(a)). Electrical power shall be provided through the AC Null Detector Cable Assembly (3.4.2.4.7.1) from an external 60 to 400 Hz source.

3.4.2.4.3 Controls - The controls shall consist of the following:

(a) Function Switch - This switch shall be a rotary switch with the following selections.

(1) OFF - Batteries shall be disconnected in this position.

(2) BATT CHG (Battery Charge) - In this position the AC Null Detector shall draw power through the BATTERY CHARGE (3.4.2.4.6 (b)) connector to recharge the batteries, and the Indicator shall read the charging current.

(3) BATT TEST (Battery Test) - In this position the battery voltage under load shall be measured on the Indicator.

(4) DC PEAK - In this position the AC Null Detector shall function as a 0 to 3.0 Vdc voltmeter.

(5) AC NULL LOW SENS (AC Null Low Sensitivity) - In this position the AC Null Detector shall act as a low sensitivity ac voltmeter, per (3.3.2.2(b)(1)).

(6) AC NULL HIGH SENS (AC Null High Sensitivity) - In this position the AC Null Detector shall act as a high sensitivity ity ac voltmeter, per (3.3.2.2.(b)(2)).

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(b) METER AC ZERO - This control shall be used to set the Indicator (3.4.2.4.5) pointer to 0 when in the AC NULL HIGH SENS (3.4.2.4.3 (a)(6)) position before use as a high sensitivity ac voltmeter.

3.4.2.4.4 Interlock - There shall be an Interlock provided so that when the cover is on, power shall be disconnected regardless of the position of the Function Switch.

3.4.2.4.5 Indicator - The Indicator shall be a 1-1/2 inch round, sealed, panel meter. It shall have markings BATTERY VOLTAGE, 0, 3, 6 and 9V with 1V graduations, and CHARGING CURRENT, 0, 20, 40, and 60 MA with graduations every 10 MA. Accuracy shall be ±3% of full scale. An arrow shall be on the meter face indicating direction of pointer movement for null adjustments.

3.4.2.4.6 Connectors

(a) SIGNAL INPUT - The Signal Input connector, J1, shall be a coaxial connector.

(b) BATTERY CHARGE - The Battery Charge connector, J2, shall be a miniature, quick-disconnect connector per MS3122E12-3P.

3.4.2.4.7 Cables - One each of the following Cable Assemblies shall be required:

3.4.2.4.7.1 Cable Assembly, AC Null Detector - This cable shall provide a means of recharging the Batteries of the AC Null Detector. One end shall have a connector MS3126F12-3S compatible with BATTERY CHARGE connector (3.4.2.4.6(b)) and the other end shall have a polarized connector, MS24663, marked "115 V 60~ OR 400~" (~= HZ). Length shall not exceed 15 ±1 inches.

3.4.2.4.7.2 Cable Assembly, AC Null Detector, Signal - This cable shall connect system signals to the Null Detector. One end shall have a connector compatible with Receiving Waveguide Assembly, CG-3249/APG-59, connector J23. The other end shall have a type BSM connector. The cable shall be of RF195/U coax per MIL-C-17/70. Length shall be 18 ±1 inches.

3.4.2.4.8 Combination Case - The AC Null Detector shall be enclosed in a Combination Case which shall conform to MIL-T-21200.

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3.4.2.4.8.1 Cover - Provision shall be made in the cover to store the cable assemblies of (3.4.2.4.7). Polyurethane cushioning material, charcoal gray, shall be provided in accordance with MIL-P-26514, Type I, Class 2.

3.4.3 Test Set Configuration

3.4.3.1 AN/AWG-10 Test Set - This Test Set, Missile Control Group OQ-50/AWM-21, shall be used with the AN/APG-59 Radar Set and shall be comprised of one ME-373/AWM-21 Meter, Fault Isolation (FIM) and one AC Null Detector DT-345A/AWG-10.

3.4.3.2 AN/AWG-11 Test Set - This Test Set, Missile Control Group OQ-51/AWM-21, shall be used with the AN/APG-60 Radar Set and shall be comprised of one ME-374/AWM-21 Meter, Fault Isolation (FIM) and one AC Null Detector DT-345A/AWG-10.

3.4.3.3 AN/AWG-12 Test Set - This Test Set, Missile Control Group OQ-52/AWM-21, shall be used with the AN/APG-61 Radar Set and shall be comprised of one ME-375/AWM-21 Meter, Fault Isolation (FIM) and one AC Null Detector DT-345A/AWG-10.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine compliance with all the applicable requirements.

(a) Preproduction Tests
(b) Acceptance Tests
(c) Life Tests

NOTE: Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(a) Contractor's Demonstration Tests
(b) Service Approval Tests

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4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of (4.5). The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration test and when requested by the procuring activity the equipment shall be delivered to the Government for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental testing in accordance with procedures of MIL-T-5422 with test limits modified to correspond with the limits specified herein and interference tests in accordance with MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction models shall be by the Naval Air Systems Command upon satisfactory completion of all tests. All preproduction models, except those consumed in testing or those rendered irreparable for their intended use, shall be returned to the Contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction models shall not be considered as equipments under this contract; however, such models may be reworked and submitted for acceptance as production equipments. Incorporation of design changes, in delivered production equipments, found to be necessary as a result of the preproduction test program, shall be the subject of separate and independent negotiations on an individual basis.

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4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(a) Individual Tests

(b) Sampling Tests

(c) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(a) Examination of Product

(b) Operational Test

(c) Reliability Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material, workmanship, and interchangeability requirements have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.3 Reliability Test - Each equipment shall be operated for a period of 6 hours without failure, under the conditions specified herein. Only those adjustments will be permitted which can be made by using such controls and adjustments as are accessible to the operator during the normal use of the equipment.

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an input signal from the oscillator

Temperature	Room Ambient
Humidity	Room Ambient
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum double amplitude of 0.036 inch.

The equipment shall be vibrated (without vibration isolators between the equipment and test fixtures) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions, the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be cycled periodically through its phases of operation. Should a failure occur, it should be repaired and tests started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.3.2 Sampling Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected for Sampling tests:

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
11 to 100	2 additional
101 to 200	1 additional
Over 200	1 for each additional 200 or fraction thereof

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4. 3. 2. 1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests after successful completion of the tests of (4. 3. 1).

(a) Complete operational tests at ambient room conditions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

(b) Repeat the test of 4. 3. 2. 1(a) using the highest and lowest specified input voltages and frequencies.

(c) Operational test at environmental conditions when the results from the preproduction test indicate that these environmental conditions cause degradation(s) which result in marginal or unsatisfactory performance.

4. 3. 3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4. 3. 3. 1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(a) On an early production equipment.

(b) On an equipment midway of the production run.

(c) On an early equipment after an extensive engineering change.

(d) When failure reports or other information indicate additional equipment should be tested. (This will be determined by the procuring activity.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory control. The maximum number of special tests shall be defined by the procuring agency at the time of request for bids.

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4.3.3.2 Scope of Tests - Special tests shall consist of such tests as are selected by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted:

(a) Temperature - Altitude Tests

(b) Vibration Tests (90 minutes cycling in each of 3 planes)

(c) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(a) Determine the cause of failure.

(b) Determine if the failure is an isolated case or design defect.

(c) Where practical, include a test in the individual test to check all equipment for this requirement until assurance is obtained that the defect has been satisfactorily corrected.

4.4 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in (4.4.1). The life tests sample shall be selected by the Government Inspector in accordance with the following: (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
26 to 200	1 additional
201 to 500	1 additional
Over 500	1 for each additional 500 or fraction thereof

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4.4.1 Test Conditions - The life test shall be conducted under the following conditions:

(a)	Temperature	Room Ambient
(b)	Altitude	Normal Ground
(c)	Humidity	Room Ambient

4.4.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off and put through its various phases of operation.

4.4.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.4.4 Reconditioning of Life Test Samples - An equipment which has been subject to the life test shall be reconditioned as follows:

(a) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(b) After reworking, the contractor shall resubmit the equipment for acceptance.

4.5 Test Procedure - Preproduction tests, acceptance tests and life tests shall be performed in accordance with a test plan and procedures mutually agreeable to the procuring activity and the contractor. The test plan shall be prepared by the contractor. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts, the procuring activity will furnish the procedures to be used for testing.

4.6 Presubmission Testing - No item, part, or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply to the best of his knowledge and belief, with all applicable requirements.

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4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and be resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.8 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability or interchangeability of maintenance parts of assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes and Material Bulletins shall be prepared in accordance with MIL-T-23336.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

6. NOTES

6.1 Ordering Data - Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, Number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1).

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specifications.

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(b) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Notation of Revisions - A bar in the margin denotes a change made in that paragraph from the previous version of the document. This has been done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

6.4 MTBF Basis - The MTBF specified in 3.2.2.3 is based on a relay duty cycle equivalent to the processing of no more than 200 frames in a time period of one hour as registered on the Elapsed Time Meter. Additionally, the MTBF specifically excludes the Film-Tape as a component.

6.5 Availability of Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

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SPECIFICATION ANALYSIS SHEET

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Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-T-81741(AS) TEST SET, MISSILE CONTROL GROUP OQ-50/AWM-21, OQ-51/AWM-21, OQ-52/AWM-21

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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